

Eric Han

Third-Year Mechanical & Aerospace Engineering Student | Applied R&D, Manufacturing, System Design
erichanx1226@gmail.com | 408-960-4503 | Santa Clara, CA | [Portfolio](#) | [LinkedIn](#)

EXPERIENCE

Research Assistant

Dixon Lab, Case Western Reserve University
Sep 2025 – Current

- Conducting CFD analysis in ANSYS Fluent comparing planar slicing and auxiliary control-surface integration methods for olfactory airflow quantification in identical 3D nasal airway geometries.
- Designed a multi-layer microfluidic MAqCI 'lab-on-a-chip' device in AutoCAD, resolving micron-scale channel-spacing and functional constraints through iterative design to produce a manufacturable drawing
- Developed a workflow in 3D Slicer to convert MRI scans of brains to STL CAD files for fabrication of brain matrices used in dissection-related research

Operations Engineering Intern

Belden Inc., Washington PA
May 2025 – Aug 2025

- Led end-to-end design of an industrial rack assembly for a powder coat line using Inventor and AutoCAD, performing FEA-based structural integrity verification and Excel-based cost-benefit projections
- Deployed rack reduced part loading time by **60%** while improving space utilization and heat uniformity
- Documented and standardized 100% of the automatic powder-coating line by creating an ISO 9001:2015 standard operating procedure
- Explored newly available Python-Excel integration capabilities and presented use cases at operations and HQ meetings, generating cross-team interest in automation and machine learning opportunities

Mechanical Member

Case Western Reserve University Motorsports, Cleveland OH
Jan 2024 – Jan 2025

- Machined parts by hand using mill, lathe, saw, etc. and performed vehicle maintenance tasks
- Worked with designs and part drawings in Siemens NX and SolidWorks

EDUCATION

BSE, Mechanical/Aerospace Engineering – 3.67 GPA

Expected Graduation: May 2027

Case Western Reserve University, Cleveland OH

PROJECTS

Brain Matrix (Dixon Lab)

- Create dissection-use brain matrix via 3D printing as a much more cost-effective alternative to purchasing
- Designed a two-part matrix by embedding the inferior brain geometry into a mesh block and projecting the top view of the cut upwards to define the upper half
- Fabricated and evaluated two PLA prototypes via Ultimaker FDM printing, transitioning to resin printing for improved dimensional precision and surface finish

Airfoil Analysis

- Tested NACA 2415 airfoil in subsonic wind tunnel from -10° to $+20^\circ$ angle of attack (5° increments)
- Used MATLAB model to convert raw pressure data into a pressure coefficient map for sixteen points across the airfoil surface, allowing computational estimates of C_p , C_L , and C_D

SKILLS

- Mechanical – SolidWorks (CSWA 2023), AutoCAD, Inventor 3D, Siemens NX, FEA, ANSYS Fluent
- Computer – Excel, Python, Java, MATLAB, Data Analysis & ML: Regression, Classification
- Machining & Fabrication – Mill, Lathe, Drill Press, Waterjet, 3D Printer